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unit 1 - Introduction to Computer and Robotics Hardware

PDF Documents

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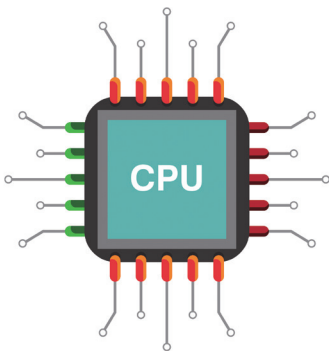
UNIT 1 INTRODUCTION TO COMPUTER AND ROBOTICS HARDWARE

Our daily lives are filled with technology – from the smartphones and tablets that connect us to the world, to the robotic toys and devices that entertain and assist us. But have you ever stopped to think about what goes on inside these gadgets? How does pressing a button on a game controller bring to life the characters on your screen? Or how can a robot vacuum navigate around your room, avoiding obstacles and keeping things clean?



Just like the stories in your favorite books or movies, there's a whole world inside every computer and robot, filled with tiny components working together to perform incredible tasks. These components are the unsung heroes of technology, each playing a crucial role in bringing our digital experiences to life.

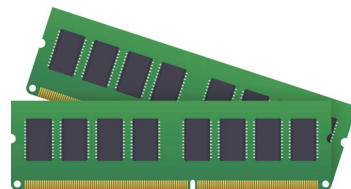
1. CPU (CENTRAL PROCESSING UNIT)



Think of the CPU as the brain of the computer and robots. Just like you use your brain to think, solve puzzles, and make decisions, the CPU

uses its "brainpower" to follow instructions from games or programs. It's incredibly smart and can do millions of calculations every second to make sure everything runs smoothly.

2. MEMORY (RAM AND STORAGE)

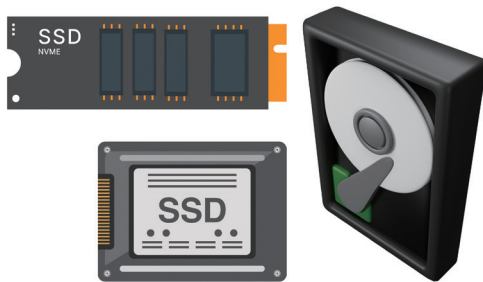


a. Ram

Imagine you're working on a puzzle. You keep some pieces in your hands because you need them soon – that's what RAM does for a computer. It holds onto information that the computer needs right away, making everything you do, from playing games to writing stories, super fast.

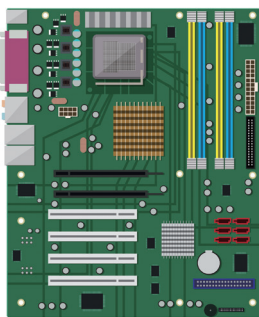


b. Storage



Now, think about your toy box, where you keep toys for when you want to play with them later. Long-term storage, like hard drives or SSDs, is the computer's toy box, storing all sorts of things it knows, from your favorite photos to the games you love, safe and sound until you need them.

3. MOTHERBOARD



The motherboard is like the foundation of a house or the halls of a school, connecting all the different parts. It's where the CPU, memory, and other components "meet" and "talk" to each other.

Without the motherboard, the computer wouldn't work because nothing would be connected!

4. INPUT DEVICES (KEYBOARD, MOUSE)



These are how we tell the computer or robot what we want it to do. When you press a key, click a mouse, or even speak into a microphone, you're using input devices to send information to the computer, just like when you raise your hand in class to give an answer.

5. OUTPUT DEVICES (MONITOR, SPEAKERS)



Output devices are how the computer or robot shows us what it's doing or what it has found out. When you look at a monitor and see your game or hear music from the speakers, the computer is using output devices to share information with you, just like when your teacher writes on the board to show you something new.

6. SENSORS AND ACTUATORS (ROBOTICS FOCUS)

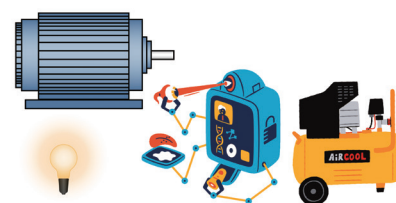
a. Sensors

Robots use sensors to understand what's happening around them.



Sensors can help a robot "see" with a camera, "hear" with a microphone, or "feel" with touch sensors. It's how robots get to know their world, just like you use your senses to learn and explore. These all function in similar ways to input devices of a computer.

b. Actuators



Actuators are the parts that let robots move or interact with things. They can be motors that

help a robot walk, wheels that let it roll, or arms that allow it to pick up objects. Actuators are like the muscles that let you run, jump, and play. You can consider actuators in robotics "output devices" in a computer.

Activity: "My Body: Inputs and Outputs"

INSTRUCTIONS:

Look at the provided image with the outline of a human body and zoomed-in images of specific body parts: eyes, ears, mouth, hands, and feet. Then, identify the function of each body part in the context of input, output, and dual-function.

